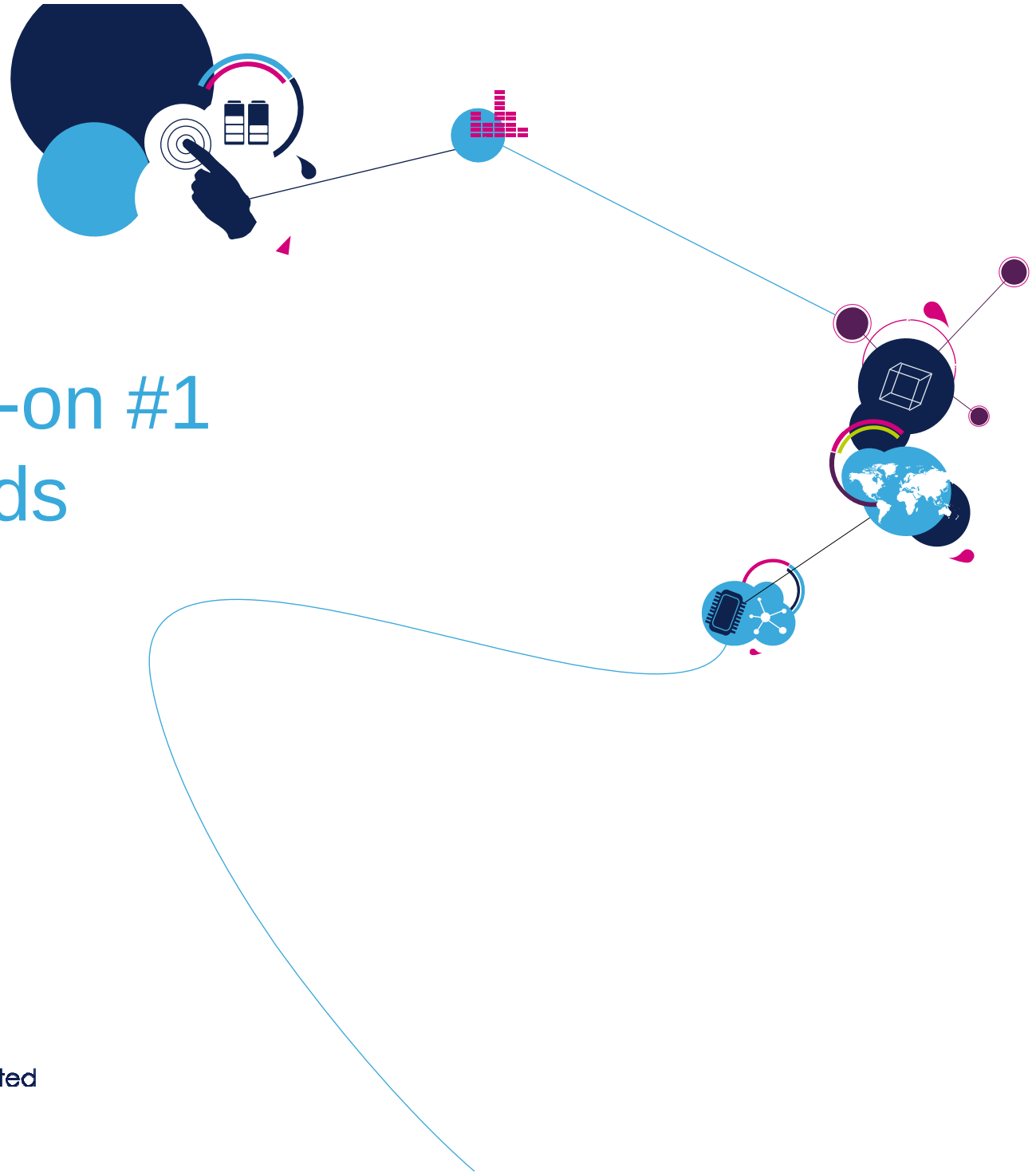


LoRa hands-on #1

AT commands

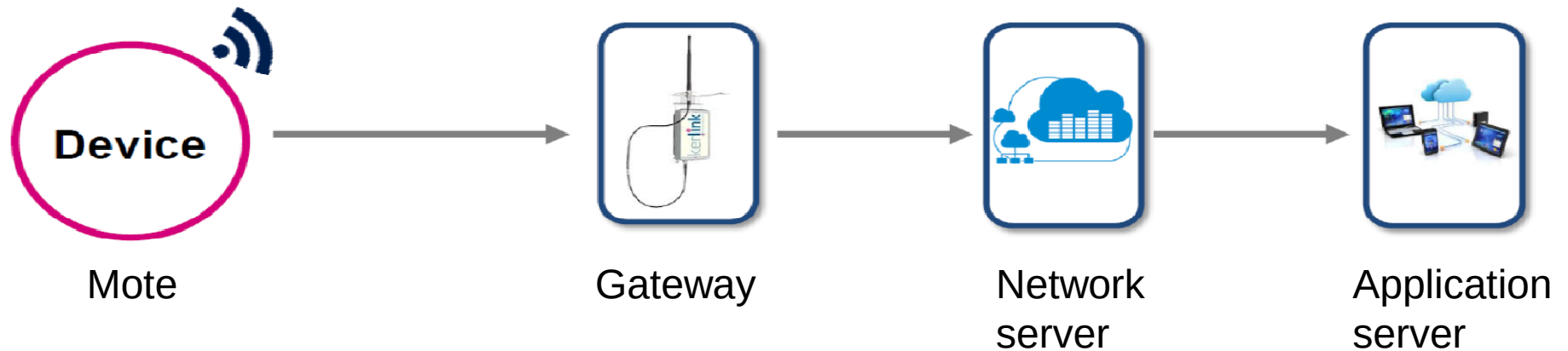


Hands-on

- Key learning
 - First touch with LoRa technology
 - What does LoRa network consist of ?
 - How to send and receive data using LoRa class A end node modem ?

Hands-on

How to establish connection in 3 steps using AT commands driven LoRa modem

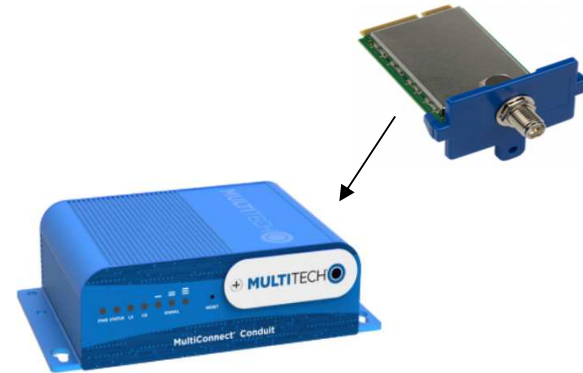


What do we need ?

1. LoRa gateway (facilitator only).

One of possible solution: MultiConnect Conduit,

- embeds both LoRa gateway and network server,
- TCP/IP connection possible via Ethernet or GSM module,
- WEB user interface.



What do we need ?

2. LoRa mote.

B-L072Z-LRWAN1 discovery kit,
-embeds the CMWX1ZZABZ-091 LoRa
module with **STM32L072** MCU inside,
-includes ST-LINK/V2-1,
- antenna connectors,
- Arduino connector,
- LEDs, push-buttons, USB OTG.



Detailed description: UM2115 (DocID029802 Rev 1)

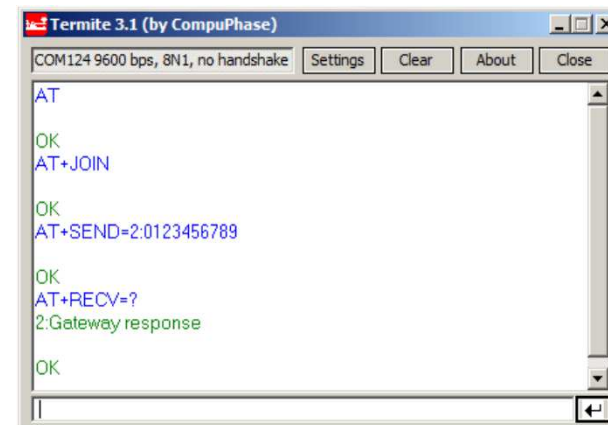
Hands-on

What do we need ?

3. AT command driven LoRa mote firmware.

(I-CUBE-LRWAN software package, “AT_Slave” project)

4. PC terminal software



Hands-on

Step 1

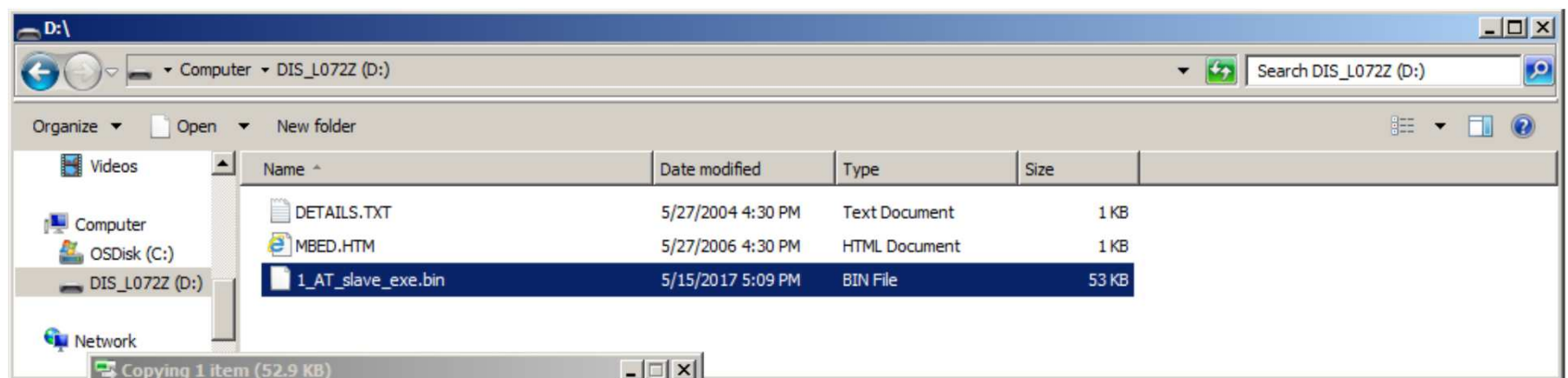
Download firmware to B-L072Z-LRWAN1 discovery board:

a) connect B-L072Z-LRWAN1 board to PC USB port via micro USB cable

b) Start Windows Explorer and open ST-Link drive

c) drag and drop

“...Hands-on_framework\Projects\Multi\Applications\LoRa\AT_Slave\1_AT_slave_exe.bin”
firmware file to ST-Link drive

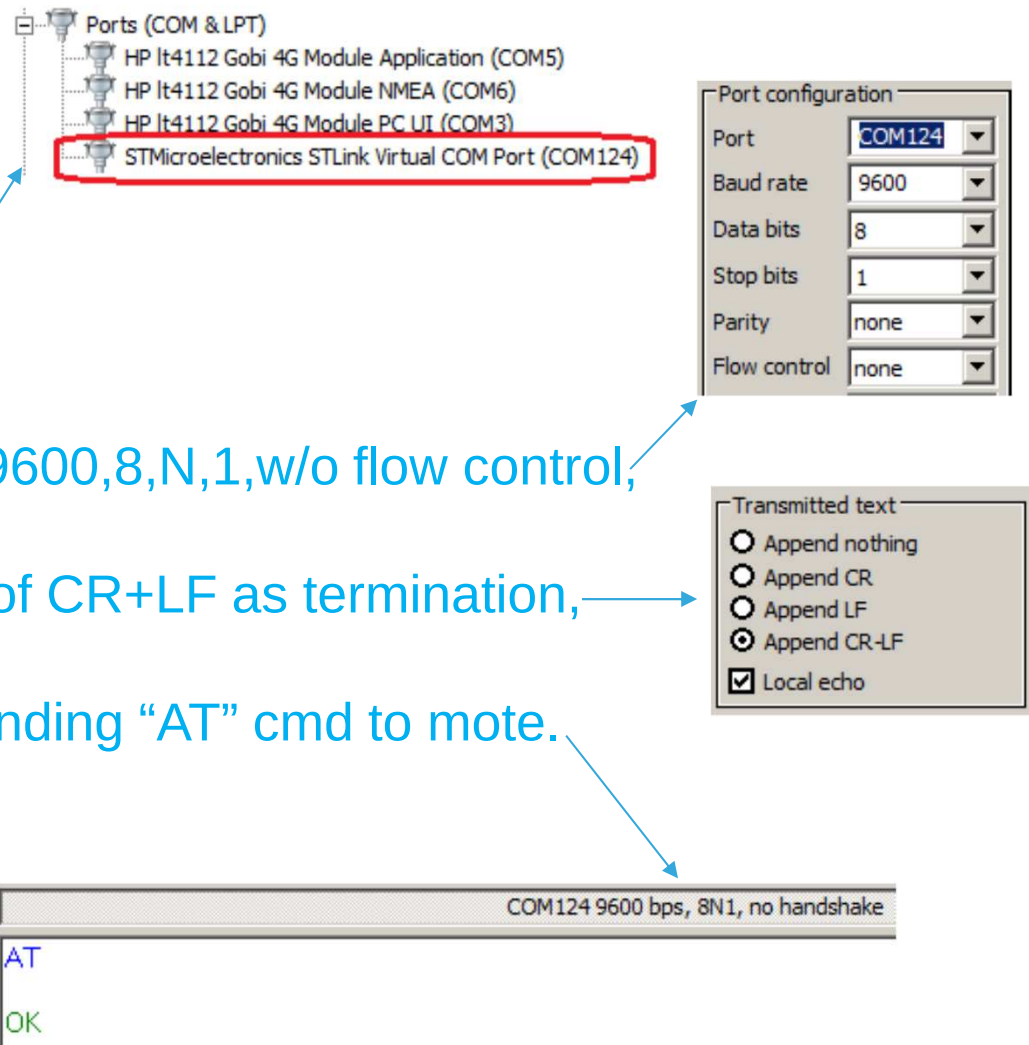


Hands-on

Step 2

Start terminal application:

- select proper COM port,
- set COM port parameters: 9600,8,N,1,w/o flow control,
- set local echo and append of CR+LF as termination,
- verify communication by sending “AT” cmd to mote.



Hands-on

Step 3

Send and receive data using AT commands:

- a) join to the network, **AT+JOIN**
- b) check network join status, **AT+NJS=?**,
- a) if join status is **1** then send data to gateway, **AT+SEND=2:MOTE_Name_Surname**
- b) After few seconds, check if gateway response is received, **AT+RCV=?**

LoRa gateway response

```
COM124 9600 bps, 8N1, no handshake
AT
OK
AT+JOIN
OK
AT+NJS=?
0
OK
AT+NJS=?
1
OK
AT+SEND=2:MOTE_Name_Surname
OK
AT+RCV=?
0:
OK
AT+RCV=?
2:GWRESP_Name_Surname
OK
```

Detailed AT commands description: AN4967

(DocID030084 Rev 1)

Hands-on

Thank you